

Metrea Aerospace Design

When to use a DER for Alterations

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Today's Team

TJ Spitzmiller
Pipeline Development Director

Laura Wilding
ODA Certification Manager

Shilah Chhadua
Aircraft Certification Engineer



When a modification to an aircraft is considered major or complex!

What, no STC? OK, 14 CFR Part 43 it is!

- The DER approval is for engineering design data only and is not an installation approval.
 - This statement is not intended to prevent the data from being used as the basis for a major alteration, but to emphasize to the installer their responsibility to perform the alteration with all the required approved data.
- The DER 8110-3 is only part of the solution
- The installer performs alteration with required approved data
- Form 337



What is a Designee?

A private person or organization delegated to act as a representative of the Administrator.

Both individuals and organizations are considered designees, and the FAA is responsible for overseeing and managing them.

The FAA doesn't have the resources to manage the growing aviation industry. By using designees for routine certification tasks the FAA can focus its limited resources on safety and new technologies.



DER Technical Disciplines

- Electrical
- Structural
- Mechanical
- Powerplant
- Acoustical

- Systems
- Flight Test/Analyst
- Engine
- Propeller
- Flammability



DER Responsibilities

- Determine Cert Basis
 - Set of airworthiness and environmental requirements that must be met
- Approve Test Plans
- Handle Non-Conformities
 - Deviation from policy, procedure or requirement
- Test Witnessing
- Approve Tech Data
- Review and Approve Descriptive Data
 - Drawings, analysis, photos, and specifications

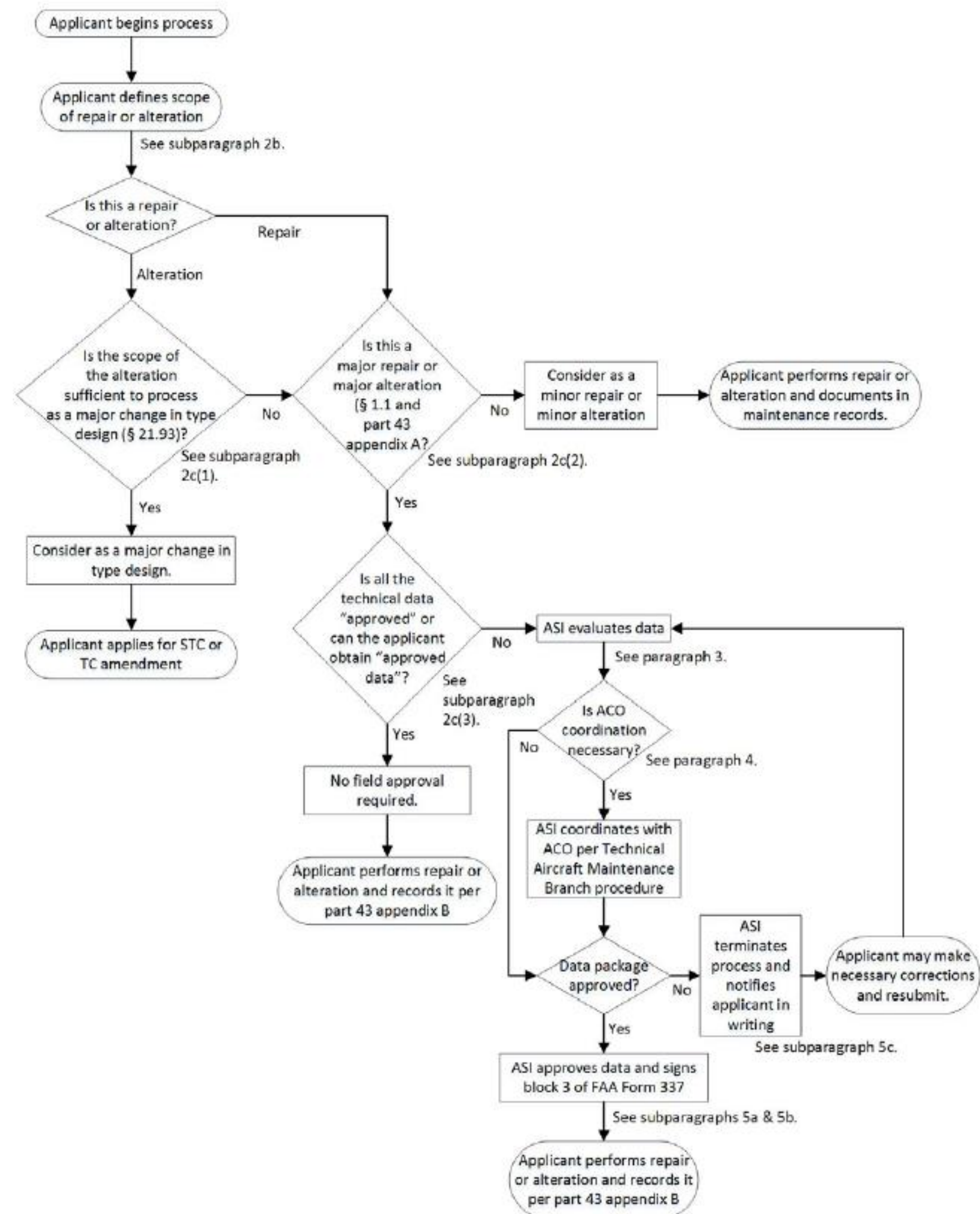
ONLY by issuing FAA Form 8110-3



Order 8300.16A

Major Repair & Alteration Data Approval

- Defines the process for the approval of technical data for major repairs and major alterations.
- Does not provide guidance for making decisions on major or minor changes in type design.
 - Audience includes: FSDO, Cert Office, DER, ODA, DAR and others engaged in the approval process for major repairs, major alterations, and the data associated with each.



Major Repair & Alteration Job Aid

AFS-300 MAJOR REPAIR/ALTERATION JOB AID

Figure 4-68 Order 8900.1 Volume 4, Chapter 9, Section 1
TRANSPORT CATEGORY AIRPLANES

	Systems (TCAS) I. (See InFO 08047)		
9f	TCAS II	STC	INFO
9g	Autopilot, Flight Guidance Systems, and automatic flight control systems (AFCS) or flight directors (FD)	STC	
9h	Cockpit voice recorders (CVR) and associated interfaces.	ENG	
9i	Flight data recorders (FDR) and associated sensors or digital flight data acquisition unit (DFDAU)	STC	
9j	Electronic flight instrument systems (EFIS).	STC	
9k	Electronic Horizontal Situation Indicator (EHSI). (See AC 25-11 and InFO 08047.)	EVL	INFO
9l	Ground proximity warning systems (GPWS)	ENG	INFO
9m	Terrain Avoidance and Warning Systems (TAWS) - A	ENG	INFO
9n	TAWS-B. (See AC 25-23 and InFO 08047.)	ENG	INFO

The Major Repair and Alteration Data Approval Job Aid lists alterations that the FAA has determined must be processed as major changes in type design.

Guidance

- **AC 43-210A**
 - This advisory circular (AC) describes a standardized procedure for requesting approval of technical data associated with major repairs/major alterations.
- **AC 43.13-1B**
 - This advisory circular (AC) contains methods, techniques, and practices acceptable to the Administrator for the inspection and repair of non-pressurized areas of civil aircraft, only when there are no manufacturer repair or maintenance instructions
 - The repairs identified in this AC may only be used as a basis for FAA approval for major repairs
- **AC 21-47**
 - This advisory circular (AC) provides guidance for creating a complete data package when data is submitted to a Federal Aviation Administration (FAA) aircraft certification branch, or to FAA designees, to support a major repair or major alteration

Acceptable means, but not the only means.



Accepted Approvals

- **Field Approval**
 - Method by which the FAA approves technical data for a major repair/major alteration on a single aircraft, and gives operational approval by reviewing and approving an AFMS
- **Approved Engineering data Package**
 - Major alterations and major repairs that have all the necessary DER- or ODA-approved technical data should not require further approval.



Thank you

Any Questions?

